

EXPERIMENT-2

Explore Git and GitHub

Task-1

ReadMe Tasks

Task 1: Professional Student Portfolio README

1. Create a GitHub repository named Student-Portfolio.
2. Create a README.md file.
3. Add your introduction, education, skills, projects, certifications, and contact information.
4. Upload your profile image and add GitHub and LinkedIn links.
5. Use the following Markdown syntax: Headings, Bold & Italic, Lists, Tables, Images, Hyperlinks, Badges, Horizontal Lines, and Emojis.

Task 2: Programming Snippet Library

1. Create a repository named Programming-Snippets.
2. Create folders for Python, Java, and JavaScript.
3. Add at least three programs in each folder.
4. Create a README.md linking to every program.
5. Use the following Markdown syntax: Headings, Lists, Internal Links, Code Blocks, Tables, Badges, Inline Code, Block Quotes, and Task Lists.

Task 3: Mini Project Documentation

1. Choose any software project.
2. Create a README.md file.
3. Include project overview, objectives, features, technologies used, folder structure, installation steps, screenshots, and future scope.
4. Use the following Markdown syntax: Headings, Images, Tables, Code Blocks, Badges, Hyperlinks, Horizontal Lines, Comments, and Emojis.

Task 4: DevOps Tools README

1. Create a repository documenting Git, Docker, Jenkins, Kubernetes, and AWS.
2. For each tool include its purpose, official website, one commonly used command, and a badge.
3. Present the content neatly.
4. Use the following Markdown syntax: Headings, Tables, Hyperlinks, Code Blocks, Badges, Lists, Inline Code, Images, and Block Quotes.

Task 5: GitHub Profile README

5. Create a repository with the same name as your GitHub username.
1. Create a professional README.md.
2. Include introduction, skills, GitHub statistics, visitor counter, streak statistics, top languages, social media links, and contact details.
3. Use the following Markdown syntax/features: Headings, Badges, Images, Animated GIF, Typing Animation, Visitor Counter, GitHub Statistics, GitHub Streak, Top Languages, GitHub Trophy, Activity Graph, Animated Banner, Animated Footer, Social Media Badges, and Emojis.

Collaboration:

Invite classmates as collaborators using the **Settings > Collaborators** tab. Use **pull requests** to review and merge contributions.

Task-2

Task:Git Branching Story Project: "A Tale in Branches"

Use Git branches to create a collaborative or personal story. Each branch represents a chapter, and finally, they merge all into a single book.html file.

Step 1: Create Repository on GitHub

1. Sign in to GitHub.
2. Click + → New repository.
3. Repository name: stories
4. Select Public.
5. Do NOT initialize with README, .gitignore or License.
6. Click Create repository.
7. Copy the repository URL.

Step 2: Create Local Project

```
mkdir stories
cd stories
git init
git branch -M main
```

Step 3: Create index.html

```
nano index.html
```

Add a simple HTML page, save using Ctrl+O, Enter, Ctrl+X.

Step 4: First Commit

```
git add index.html
git commit -m "Initial commit"
```

Step 5: Connect to GitHub

```
git remote add origin https://github.com/YOUR-USERNAME/stories.git
git remote -v
git push -u origin main
```

Step 6: Create Branch chapter1

```
git switch -c chapter1
# edit index.html
git add index.html
git commit -m "Add Chapter 1"
git push -u origin chapter1
```

Step 7: Merge chapter1

```
git switch main
git merge chapter1
git push origin main
```

Step 8: Create Branch chapter2

```
git switch -c chapter2
```

edit index.html

```
git add index.html
git commit -m "Add Chapter 2"
git push -u origin chapter2
```

Step 9: Merge chapter2

```
git switch main
git merge chapter2
git push origin main
```

Step 10: Verify Repository

```
git branch -a
git log --oneline --graph --all
git status
```

Step 11: View Files Locally

Open C:\Users\<username>\stories and double-click index.html or run:
start index.html

Step 12: Publish Using GitHub Pages

1. Open repository on GitHub.
2. Go to Settings → Pages.
3. Source: Deploy from a branch.
4. Branch: main
5. Folder: /(root)
6. Save.
7. After deployment, open:
<https://YOUR-USERNAME.github.io/stories/>

Expected Result

The browser displays the HTML page containing the initial story, Chapter 1, and Chapter 2. GitHub shows the branches main, chapter1, and chapter2 with the complete commit history.